MILITARY SPECIFICATION SHEET

MULTIPLEXERS, DEMULTIPLEXERS, MULTIPLEXERS/DEMULTIPLEXERS (MULDEMS), MULTIPLEXER, FREQUENCY-DIVISION, FIBER OPTIC, SHIPBOARD, 4:1 CHANNELS, 30 MHZ (DOUBLE SIDEBAND) (METRIC)

This specification is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-M-24731.

CLASSIFICATION

Equipment function: M (frequency-division multiplexer).
Signal type: B (Optical).
Channel configuration: A (1 to 16 channels).
Analog subchannel class: F (Not greater than 30 MHz).
Digital subchannel class: G (Not greater than 100 Mbps).
Style: C (Vertical rack plug-in module).

RELIABILITY: MTBF greater than 10,000 hours.

DESIGN AND CONSTRUCTION

Enclosure: Watertight.
Fiber optic transmitters: The transmitter shall operate at 1.30 micrometer (μm) with multimode fiber. The transmitter output power shall be 0.1 milliwatt (mW).
Fiber optic receivers: N/A.
Fiber optic pigtail: In accordance with MIL-C-85045 and MIL-C-85045/14.
Fiber optic connectors: N/A.

Dimensions:

Mainframe assembly: See figure 1.
Multiplexer module: See figure 2.
Fiber optic transmitter module: See figure 3.
Power supply module: See figure 4.

FSC 6008
AMSC N/A
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Weight:

Mainframe assembly with power supply: 6.8 kg (max.).
Multiplexer module: 1.1 kg (max.).
Fiber optic transmitter module: 1.6 kg (max.).

GENERAL PERFORMANCE CHARACTERISTICS

Multiplexers: 4 input channels, 1 output channel; 70 MHz per input channel.

Subchannel input interface element: The subchannel interface element shall accept baseband information carried on an intermediate frequency (IF) of 70 MHz and not exceeding a 3-dB double-sideband bandwidth of 30 MHz. The 70-MHz IF carrier shall be frequency translated according to the spacing plan of: f_{out} = 52.5 MHz + (N-1)X35 MHz, where N = number of channels.

Combiner/amplifier element: passive 4-way combiner.

Multiplexer output interface element: The output of the combiner shall be compatible with the input signals for MIL-T-24735. The multiplexed output signal of the multiplexer output interface element shall be an optically modulated signal in accordance with MIL-T-24735.

DETAILED PERFORMANCE CHARACTERISTICS

Operational performance parameters:

Prime power and current load for multiplexer:

5 ± 0.25 volts	at 240 milliamperes (mA) max.
15, minus 0.75, plus 2.25 volts	at 300 mA max.
Minus 15, minus 0.75, plus 2.25 volts	at 35 mA max.

Prime power and current load for transmitter unit:

Minus 15, minus 0.75, plus 2.25 volts	at 100 mA max.
15, minus 0.75, plus 2.25 volts	at 5 mA max.
5, minus 0.25, plus 0.75 volts	at 1.4 amperes max.

Ac prime power: 120 volts, 60 hertz (Hz), 7.5 amperes max.

Alarm and fault conditions: Status monitoring for power supplies, input signal voltage level, and transmitter output levels.

Alarm indication signal: N/A.

Modulation formats: Frequency modulation in accordance with the frequency plan.

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Guard-band width: 5 MHz.

Propagation delay: 10 nanoseconds.

Pilot tone characteristics: N/A.

Frequency plan: 52.5 to 577.5 MHz in 35-MHz steps, with a 3-dB double-sideband bandwidth of 30 MHz.

Interchannel crosstalk: N/A.

Optical performance:

Dynamic range: 20 dB.

Channel performance parameters:

Analog subchannels: The IF input signals shall have the following characteristics:

Impedance: 75 ohms.
Return loss: 20 dB, minimum.
Frequency: 70 MHz.
Power level: 4 dBmW ± 2 dB.

Digital channels: N/A.
Other signal types: N/A.

Interface characteristics:

Analog subchannel interface: The radio frequency (RF) output signals of the multiplexer module shall have the following characteristics:

Impedance: 75 ohms.
Return loss: 15 dB, minimum.
Frequency range: 52.5 to 577.5 MHz in 35-MHz steps.
Power level: minus 1 to plus 7 dBmW (adjustable).
Frequency stability: plus or minus 50 kHz.
Frequency accuracy: plus or minus 50 kHz.

Digital channels: N/A.

Multiplexed signal interface:

FDM signal output:

Electrical interface: N/A.
Optical interface: Optically modulated signal in accordance with output signals of MIL-T-24735.

FDDM signal input: N/A.

Maintenance orderwire: N/A.
Other interfaces: N/A.

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ENVIRONMENTAL REQUIREMENTS

Temperature range: 1 (operating - 0 to plus 85°C;
storage - minus 62 to plus 85°C).
Nuclear radiation resistance: N/A.

ELECTROMAGNETIC INTERFERENCE SUPPRESSION

Electromagnetic: Below-deck application, in accordance with MIL-STD-461, part 5.

Conducted susceptibility: CS01, CS02, CS06, CS09.
Conducted emission: CE01, CE03, CE06.
Radiated emission: RE01, RE02, RE03.
Radiated susceptibility: RS01, RS02, RS03.

Lightning: Transient suppression for voltage spikes of 2500 volts.
Power leads: N/A.

SAFETY

Laser safety: N/A.

METHODS OF INSPECTION

OPERATIONAL TESTS

Pilot tones: N/A

ENVIRONMENTAL TESTS

Explosive atmosphere: The values for the test conditions shall be
Fuel - n-hexane (reagent grade or 95 percent).
Fuel amount - 3.8 percent by volume of the test chamber.
Test altitude - sea level.
Test temperature - Plus 85°C.

Electromagnetic interference suppression: Tests shall be conducted in accordance with MIL-STD-462, over a frequency range of 14 KHz to 600 MHz. The RS03 limit shall be 200 volts per meter. Tests shall be conducted at ambient conditions. Verification of operation shall be performed after the test.

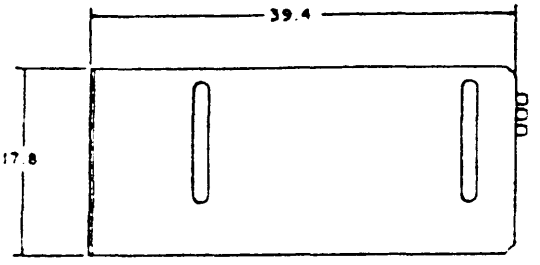
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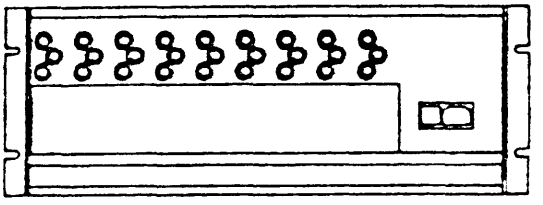
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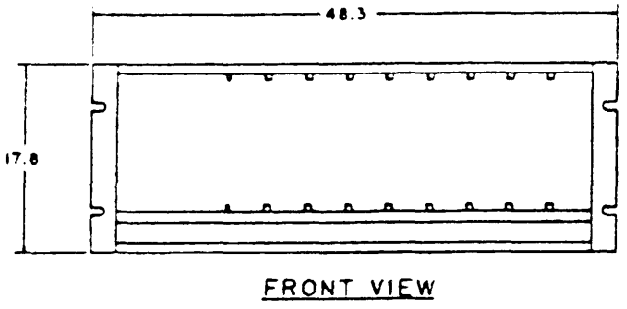
Review activities:
Navy - EC, YD

Preparing activity:
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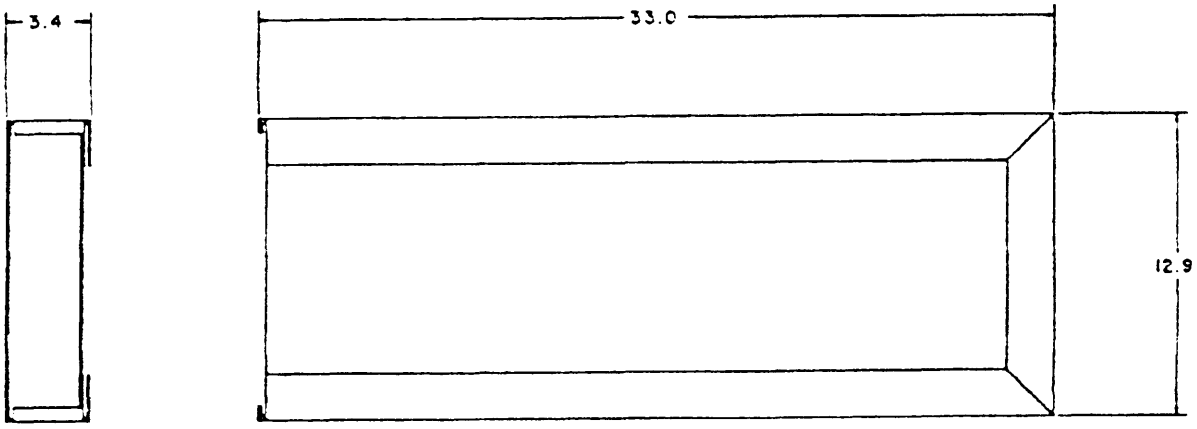




NOTE:
1. All dimensions are in centimeters.

FIGURE 1. Mainframe assembly.

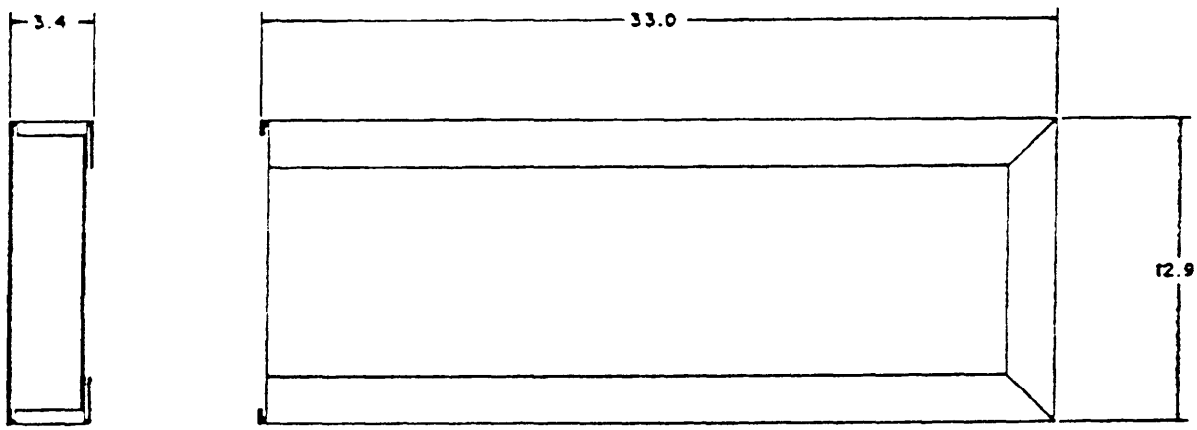
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NOTE:
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FIGURE 2. Multiplexer module.

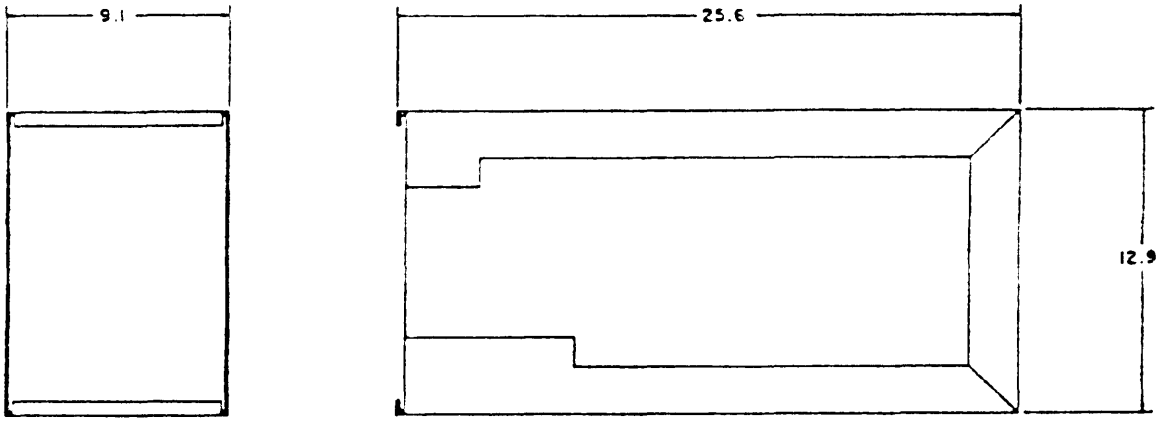
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NOTE:
1. All dimensions are in centimeters.

FIGURE 3. Fiber optic transmitter module.

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NOTE:
1. All dimensions are in centimeters.

FIGURE 4. Power supply module.

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